University Student-Support Case Agent

BSE4104 Emerging Trends in Software Engineering

Group M Evening | Final Presentation

- Grounded answers
- Bounded actions
- Human hand-off

The problem and who it is for



User

Makerere freshmen (first-year) going through onboarding



Pain

Answers scattered across documents and offices; queues for routine FAQs



Need

Grounded answers, plus a support draft when the documents are not enough

Out of scope Admissions, grading, discipline, fee decisions and auto-resolving cases

Why AI is justified, and where it stops



AI

Retrieval-grounded Q&A, plus deciding when to escalate to a draft



Deterministic software

Validation, case IDs, queue routing, limits and the consent gate



Human staff

Review and resolve every draft

AI does not invent policy, waive fees, close cases, or act without agreement.

From model to bounded agent in eight weeks

W2 Foundation model

Versioned prompts, v1.0 to v1.1

W3 RAG

Corpus, retrieve, cite, honest fallback (v1.2)

W4 Tools

retrieve_documents and create_support_case (v1.3)

W5 Bounded agent

Sense, Plan, Act, Observe, Stop

W6 State and memory

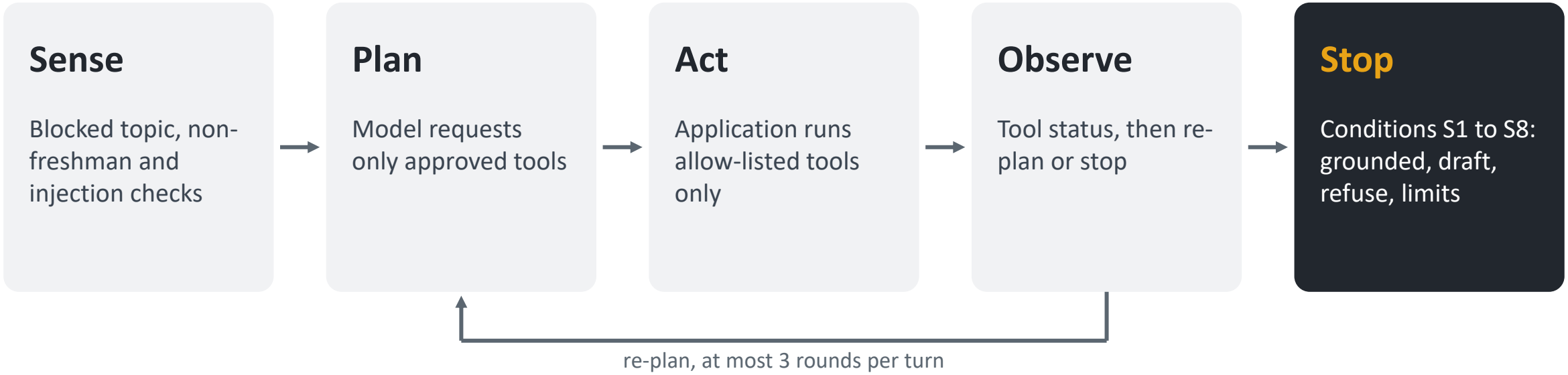
Explicit state, justified memory, MCP-style interface

W7 Evaluation

Observability, guardrails, failure catalogue

Week 8: hardened release, final report and demonstration

The bounded agent loop



Limits are enforced in application code, not by the model

3	model rounds per turn
3	tool calls per turn
1	support draft per session

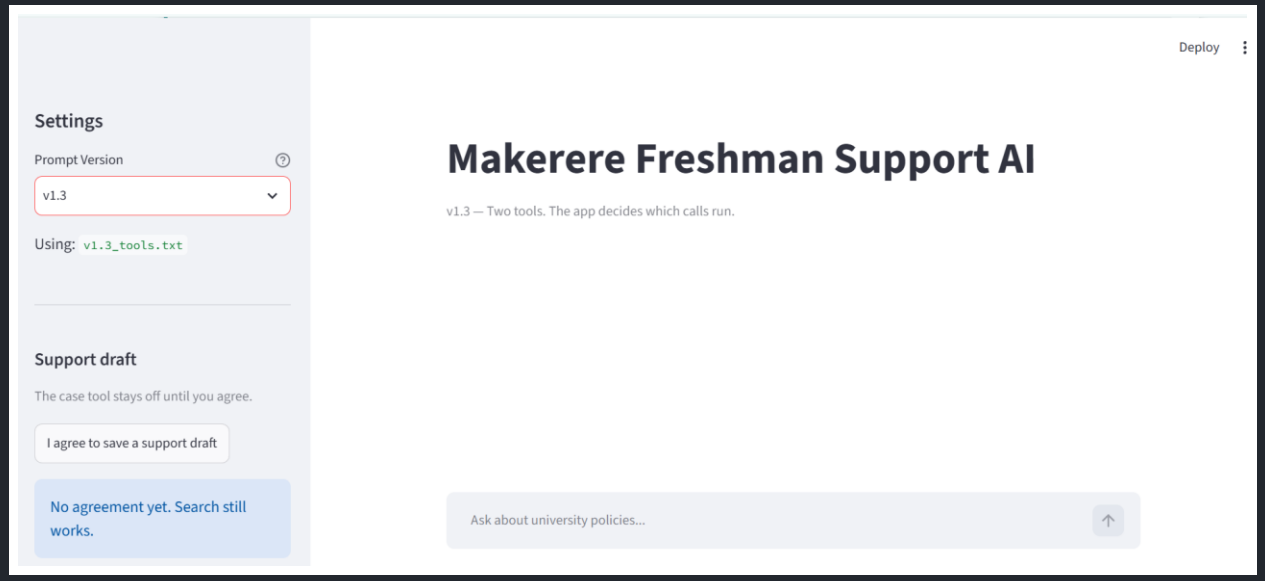
Live demo: from question to staff hand-off

1 **Ask a grounded question**

Cited source, tool used: retrieve_documents, stop condition S1

2 **Ask for staff help**

Agree in the sidebar and give a valid registration number. Case created: case_id, queue label, stop condition S2



The consent control lives in the sidebar. The case tool stays off until the student agrees.

We can inspect what the agent did

What every turn records

Tool name	which tool the model asked for
Arguments	what the call was made with
Status	ok, no_match, validation_error, created
Stop condition	S1 to S8, why the turn ended
Sources	which documents grounded the answer

10/10

guardrail checks passed (G01 to G10)

Evidence we can open

- Agent traces T1 to T3 and O1 to O4
- Guardrail screenshots G01 to G10
- Scenario evaluation results

Failure and recovery: bad input is caught

1

Student gives 23/A/93939

→

2 validation_error

The application rejects the format

→

3

The agent asks for a valid number (U or X, not A)

→

4 draft created

Retry with a valid number



Also observed

Gemini 503 (high demand): the student sees an error and can retry later

Lesson

The application owns validation. The model cannot invent a valid case.

Guardrails and human control

Agreement first

Consent is a sidebar control; the case tool stays off until the student agrees

G01

Blocked topics

Admissions, grading, discipline and fee decisions are refused

S5 | G03, G04

Scope and injection

Non-freshman requests stop at S4; override attempts stop at S6

S4, S6 | G05, G06

One draft per session

A second draft is refused and points to the existing case

G08

Tool allow-list

Only retrieve_documents and create_support_case can run

G02

Human hand-off

Every draft goes to staff; the agent never resolves or closes cases

Staff resolve

Known limitations



Pattern-based injection check

It catches the override phrasings we listed, not every adversarial wording



Small, static corpus

Eight documents; dynamic portal content such as registration deadlines is excluded



Shared free-tier API

Quota limits and 503 errors can interrupt a session

Key lessons

1. **Grounding and citations** beat unrestricted chat
2. **Orchestration must enforce limits;** do not trust the model alone
3. **Consent is a UI flag,** not a free-text yes
4. **Evaluate refusals and failures,** not only happy paths
5. **Keep scope small** enough to test, explain and control

Thank you. Questions?

Group M Evening

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BSE4104 Capstone

Repository and evidence

github.com/SumayyaN/BSE41014_capstone_group_m_eve

App: python -m streamlit run src/app.py

Traces: evidence/traces/

Guardrails: evidence/screenshots/guardrails/

Evaluation: docs/evaluation/

Reports: docs/weekly-reports/ and docs/architecture/